BAFNet: Bottleneck Attention Based Fusion Network for Sleep Apnea Detection

Xianhui Chen, Wenjun Ma, Weidong Gao, and Xiaomao Fan

***Abstract*—Sleep apnea (SA) is a common sleep-related breathing disorder that tends to induce a series of complications, such as pediatric intracranial hypertension, psoriasis, and even sudden death. Therefore, early diagnosis and treatment can effectively prevent malignant complications SA incurs. Portable monitoring (PM) is a widely used tool for people to monitor their sleep conditions outside of hospitals. In this study, we focus on SA detection based on single-lead electrocardiogram (ECG) signals which are easily collected by PM. We propose a bottleneck attention based fusion network named BAFNet, which mainly includes five parts of RRI (R-R intervals) stream network, RPA (R-peak amplitudes) stream network, global query generation, feature fusion, and classifier. To learn the feature representation of RRI/RPA segments, fully convolutional networks (FCN) with cross-learning are proposed. Meanwhile, to control the information flow between RRI and RPA networks, a global query generation with bottleneck attention is proposed. To further improve the SA detection performance, a hard sample scheme with k-means clustering is employed. Experiment results show that BAFNet can achieve competitive results, which are superior to the state-of-the-art SA detection methods. It means that BAFNet has great potential to be applied in the home sleep apnea test (HSAT) for sleep condition monitoring.**

***Index Terms*—Sleep apnea, deep neural network, ECG signals, home sleep test.**

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The source code is released at https://github.com/Bettycxh/Bottleneck-Attention-Based-Fusion-Network-for-Sleep-Apnea-Detection.

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I. Introduction

SLEEP apnea (SA) is a common sleep-related breathing disorder, whose symptom is the intermittent collapse of the upper airway caused by tongue muscle or fatty tissue [1]. Due to the loss of airflow, SA patients experience brain arousal, sympathetic activation, and decreased oxygen saturation in the blood during their sleep [2]. It is reported that SA will greatly increase the risk of developing diseases, such as pediatric intracranial hypertension [3], psoriasis [4], stroke [5], cognitive impairment [6], and daytime drowsiness [7]. Besides, SA even causes a sudden death event indirectly as it greatly enhances the risk of cerebrovascular morbidity [8], [9]. Therefore, it is of great significance to early diagnosing SA and making appropriate interventions to prevent malignant complications.

Clinically, the American Academy of Sleep Medicine (AASM) guidelines recommend that polysomnography (PSG) is the gold standard for diagnosing the SA owing to its high diagnostic accuracy [10]. However, PSG is expensive, cumbersome, and inconvenient, as well as requiring 6-8 hours of sleep recording in hospital with two professional physicians overnight monitoring [11]. Besides, PSG often induces the first-night effect caused by unfamiliar surroundings and disconformity. It greatly blocks the PSG application for a large population outside of hospitals.

Portable monitoring (PM), also named home sleep apnea tests (HSAT) and out-of-center sleep testing (OCST), is an alternative way to monitor an individual's sleep conditions conveniently at home. As reported in [10], [11], PM achieved acceptable results for screening out the SA population. Electrocardiogram (ECG), as one of the most collected physiological signals by PM, presents the voltage change of the heart beating in each cardiac cycle. ECG has a high cor-relationship with SA events due to the fact that the decreased oxygen saturation causes the irregular change of heart rate [12]. Meanwhile, the respiratory-induced ups and downs also reflected the signal change in ECG. Therefore, researchers shifted their focus to the ECG signals and proposed large amounts of ECG-based SA detection methods such as HMM-SVM [13], SE-MSCNN [14], LeNet [15], MSDA-1DCNN [16], ZF-Net [17], and so on.

These ECG-based methods achieved promising results, however, there still exist two major limitations: *1) how to strengthen the ability of feature extraction and the feature representation*;

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2) how to strengthen the ability of the model to discriminate hard samples. Noted that [14] utilized multi-scaled segments as the input for feature extraction. This scheme, whereas, generates a large number of feature maps that surely contain unrelated information and impairs the performance of the model.

In this study, we propose a bottleneck attention based fusion network called BAFNet for SA detection using single-lead ECG signals. Specifically, the proposed BAFNet mainly consists of five parts: RRI(R-R intervals) stream network, RPA (R-peak amplitudes) stream network, global query generation, feature fusion, and classifier. RRI and RPA stream networks share the identical network architecture employed for feature extraction and strengthening the feature presentation with cross-learning. To control the feature information flow between RRI and RPA networks, global query generation with bottleneck attention is proposed to generate the query vector from both RRI and RPA segments. To effectively fuse the RRI and RPA feature representation, channel-wise attention is employed. In addition, to meet up the problem of insufficient information of the target segment, adjacent segments along with the corresponding target segment are combined together in sequence as input. To further improve the SA detection performance, a hard sample technique with k-means clustering is presented. Furthermore, attention visualization and computational complexity are conducted. To sum up, our contributions can be summarized as follows:

1) To learn the feature representation of RRI/RPA signals, we employed fully convolutional networks (FCN) with cross-learning, *i.e.*, feature learning network. Take RRI stream network for example, it learns the RRI representation from RRI signal (encode) and reconstructs the corresponding RPA signal instead of RRI signal (decode), vice versa. Because RRI signals contain the information of R-peak positions in the RPA and RPA signals contain the information of RRI, which can potentially help the RRI and RPA representation learning.
2) To control the feature information flow between RRI and RPA networks, we propose global query generation with bottleneck attention. With global query vector from both RRI and RPA signals, it can effectively learn the SA-related feature information.
3) We propose a bottleneck attention based fusion network for sleep apnea detection with k-means clustering as the hard sampling technique, which can potentially help the BAFNet to discriminate the hard samples and improve the SA detection performance.
4) Experiment results show that the proposed BAFNet achieves cutting-edge SA detection performance, which reaches up to 91.25% and 100% accuracy in terms of per-segment classification and per-recording classification, separately.

The rest of this article is organized as follows. Section II introduces related work. Section III shows the details of the proposed method, including signal preprocessing, the proposed method, hard sample mining scheme, and performance evaluation metrics. Section IV illustrates the experiment setup. Section V presents the results and discussion. Section VI conclude this article.

II. Related Work

A. Home Sleep Apnea Test for SA Detection

Due to the deficiencies of PSG, HSAT is an alternative way for SA detection outside of hospitals. The main difference in the current literature based on HSAT for SA detection is the input physiological signals.

1) Nasal Signals: McCloskey et al. [18] applied a novel 2-D convolutional neural network (CNN) to a 2-D image wavelet spectrogram of the single-channel nasal signal. Wang et al. [19] first presented a global measure, the Respiratory Fluctuation Index (RFI) and regression models based on the RFI had a strong performance for obstructive sleep apnea (OSA) assessment. Wu et al. [20] introduced an enhanced frequency extraction network, which is an optimal architecture that automatically extracts sleep apnea-hypopnea events features from the spectrogram of nasal airflow signals.

2) Electroencephalogram (EEG) Signals: Jiang et al. [21] employed a short-time Fourier transform to generate the time-frequency images from single-channel EEG signals. A multi-scale parallel CNN with a mixed depth of layers and mixed sizes of convolutional filters is designed to automatically learn the features from time-frequency representations and make the SA detection. Vimala et al. [22] employed features such as energy, entropy, and variance to be imported for the classification process by using machine learning classifiers including support vector machine (SVM) with kernel functions, K-nearest neighbors (KNN), and artificial neural network (ANN). Taran et al. [23] introduced an adaptive decomposition for the detection of apnea events using EEG signals.

3) Oxygen Saturation (SpO_2) Signals: Vaquerizo et al. [24] utilized CNN to automatically detect apnea and hypopnea events from the SpO_2 raw data. Sharma et al. [25] used optimal duration-bandwidth concentrated wavelet transform to decompose the SpO_2 signals into various sub-bands (SBs), which were then fed to various supervised machine learning algorithms.

4) ECG Signals: ECG not only can be used as the input for SA detection but also can be utilized to detect cardiovascular diseases such as arrhythmia, atrial fibrillation, and so on [26], [27], [28]. Therefore, currently, many studies focused on SA detection based on HSAT using ECG signals.

Machine learning based methods: Varon et al. [29] used two novel features derived from the ECG, heart rate variability and RRI, which were fed into a least-squares support vector machines classifier, using a radial basis function kernel (RBF) kernel. Hassan et al. [30] employed a novel signal processing technique, namely tunable-Q factor wavelet transform (TQWT), and introduced random under sampling boosting (RUSBoost) to classify SA. Li et al. [31] proposed a method to detect OSA based on a deep neural network and Hidden Markov model (HMM) using single-lead ECG signals. The method utilized a sparse auto-encoder to learn features, which belongs to unsupervised learning that only requires unlabeled ECG signals. Fatimah et al. [32] separated ECG signals into frequency bands using the fast Fourier decomposition method, which would be fed into SVM for SA detection. However, due to the requirement

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of hand-crafted feature engineering, the performance of these machine learning based methods is quite limited.

Deep learning based methods: Deep learning based methods have powerful feature extraction, which can greatly improve the performance of the models. Therefore, many studies investigated effective deep learning based methods for SA detection. Wang et al. [15] proposed a modified LeNet-5 convolutional neural network with adjacent segments for SA detection using a single-lead ECG. Wang et al. [33] extracted features from RRI, which would be fed into CNN model having three convolution layers and three fully connected layers for OSA detection. Almutairi et al. [34] proposed a novel hybrid architecture consisting of CNNs and long short-term memory (LSTM) networks for the classification of OSA. Chen et al. [14] proposed a multi-scaled fusion network named SE-MSCNN for SA detection based on single-lead ECG signals acquired from wearable devices. The SE-MSCNN with the merits of quick response and lightweight parameters can be potentially embedded in a wearable device to provide a SA detection service for individuals in home sleep test. Yang et al. [35] proposed a one-dimensional squeeze-and-excitation (SE) residual group network to thoroughly extract the complementary information between heart rate variability (HRV) and ECG derived respiratory (EDR).

B. Hard Sample Mining

Hard sample mining is a general technique to improve model performance and is applied in many fields such as face detection [36], [37], person reidentification [38], [39], human action recognition [40], and so on. It is also employed in the medical data mining field. Chu et al. [41] employed an effective online hard example mining (OHEM) algorithm to train the region-based ConvNet detector, which automatically selected hard samples to improve training efficiency. Qu et al. [42] proposed a variant of OHEM named Enhance online hard example mining (E-OHEM) algorithm for external damage risk detection of transmission lines, which can not only effectively learn more hard examples but also avoid the problem of data imbalance. Shi et al. [43] presented a fire monitoring and alarm system based on YOLOv3 with OHEM to deal with the imbalance between easy samples and hard samples. Yu et al. [44] proposed loss rank mining (LRM) to fill the gap, which filtered some elements representing easy examples in final feature maps and forced detectors to concentrate on hard examples during training. Zhao et al. [45] adopted hard mining center-triplet loss (HCTL), a novel improved triplet loss, for mining hard training samples. Chen et al. [39] proposed an adaptive threshold of hard level, which can make the algorithm stay in step with the training process and alleviate the under-fitting and over-fitting problems simultaneously.

III. Methods

Fig. 1 shows the overall pipeline of SA detection in this study mainly consisting of two steps: signal preprocessing and SA modeling, which are described in detail as follows:

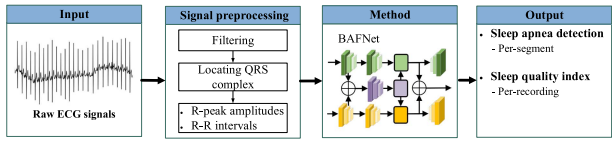


Fig. 1. Overall pipeline of sleep apnea detection using ECG signals.

A. Signal Preprocessing

Overnight ECG signals reflect the electrical activity of the heart as well as the movement of the chest. It means that ECG signals contain information related to the breathing airflow which is widely utilized for diagnosing SA events. As shown in [15], signal prepossessing can effectively depress the noise and accurately locate QRS complexes. Therefore, we follow this paradigm in this study, mainly including three steps: filtering noise, locating R-peaks, and fixing input length. ECG signals are easily contaminated by baseline wandering, powerline interference, and muscle noise. Designing an effective filter can greatly remove the noise from ECG signals. Therefore, a low-pass filter with cut-frequency ($<$16 Hz) as well as a high-pass filter with cut-frequency ($>$8 Hz) is utilized for depressing high-frequency and low-frequency noise. To obtain the SA-related RRI and RPA, locating QRS complexes is a critical step. Hence, the derivation and moving average are performed in a moving window of 80 ms to further depress noise and sharpen the R peaks. In the process of SA detection, an overnight ECG signal is divided into one-minute-long segments, generating the corresponding RRI and RPA segments. As the number of R peaks in length-fixed ECG signals may vary even if obtained from the same subject, the generated corresponding RRI/RPA segments are with various lengths. To meet the requirement of the SA detection model with fixed-length input, cubic interpolation is used to generate data points to make all the RRI/RPA segments in equal length.

Furthermore, as reported in [14], [15], adjacent segments can greatly improve the SA detection performance. Generally, a person's apnea symptoms or normal breaths usually last for a period during his/her sleep, rather than the repeated alternation between apnea and normal breathing. Therefore, adjacent segment information can, to some extent, alleviate the issue of insufficient information in a segment [15]. To leverage adjacent information, the target segment and its prior and post surrounding two segments (*i.e.*, five-minute-long sequential ECG segment) are combined together in sequence as input. And the label is marked by the corresponding target segment label.

B. The Proposed BAFNet

1) Overview: The aim of this study is to build a deep learning based SA detection model using single-lead ECG signals. As described in Section III-A, adjacent segments can meet the requirement of alleviating insufficient information of the target ECG segment to some extent. Here, the input consists of five-minute-long RRI and RPA segments generated from five-minute-long ECG signals. In order to better control the feature

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TABLE I
THE DETAILED CONFIGURATION OF THE PROPOSED BAFNET

Input		RRI	RPA
Stream network	Conv1	Conv1d(11), filters 16, stride 1, padding 5	Conv1d(11), filters 16, stride 1, padding 5
		Conv1d(11), filters 24, stride 2, padding 5	Conv1d(11), filters 24, stride 2, padding 5
		Max pooling1d(3), padding 0	Max pooling1d(3), padding 0
	Conv2	Conv1d(11), filters 32, stride 1, padding 5	Conv1d(11), filters 32, stride 1, padding 5
		Max pooling1d(5), padding 0	Max pooling1d(5), padding 0
	TConv	Upsampling1d(5)	Upsampling1d(5)
		Transpose Conv1d(11), filters 24, stride 1, padding 5	Transpose Conv1d(11), filters 24, stride 1, padding 5
		Upsampling1d(3)	Upsampling1d(3)
		Transpose Conv1d(11), filters 16, stride 2, padding 5	Transpose Conv1d(11), filters 16, stride 2, padding 5
		Transpose Conv1d(11), filters 1, stride 1, padding 5	Transpose Conv1d(11), filters 1, stride 1, padding 5
Global query generation module	Conv2	Conv1d(11), filters 32, stride 1, padding 5	
		Max pooling1d(5), padding 0	
	Self-attention		
	Cross-attention		
	Cross-attention		
Feature fusion module	Global average pooling1d (GAP)		
	Fully connected layer(48), ReLU (FC)		
	Fully connected layer(96), Sigmoid (FC)		
	Global average pooling1d (GAP)		
	Softmax(2)		

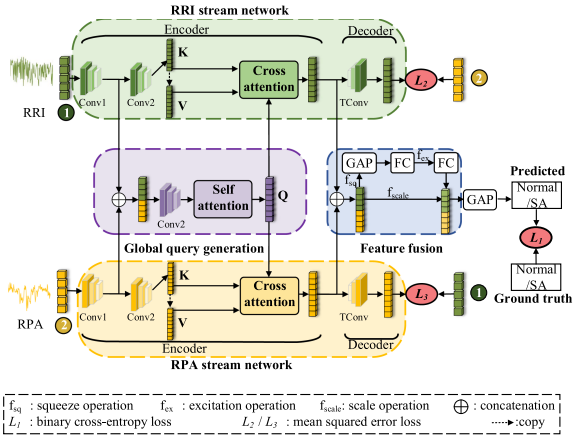


Fig. 2. Architecture of the proposed BAFNet. It mainly consists of RRI stream network, RPA stream network, global query generation, feature fusion, and classifier.

information flow among RRI and RPA stream networks, we propose a fusion network based on a bottleneck attention, which can effectively focus on the high-related feature information and precisely detect SA events.

Specifically, Fig. 2 shows the overall architecture of the proposed BAFNet. It mainly consists of RRI stream network (see in Section III-B2), RPA stream network (see in Section III-B3), global query generation module (see in Section III-B4) and feature fusion module (see in Section III-B5) followed by a global average pooling layer as well as a Softmax layer for classification. The detailed configurations of BAFNet are shown in Table I. For a brief description in the subsequent sections, we denote the input as $X = [X^{RRI}, X^{RPA}]$, where $X^{RRI} \in \mathbb{R}^d$ and $X^{RPA} \in \mathbb{R}^d$ refer to the five-minute-long RRI and RPA segments, respectively. d represents the size of five-minute-long RRI/RPA segments. Here, we empirically set the d to 900. The corresponding labels are denoted as $\mathbf{y} \in \{0, 1\}$, where $\mathbf{y} = \mathbf{1}$ refers to segments with SA events and $\mathbf{y} = \mathbf{0}$ refers to segments without SA events (*i.e.* segments with normal conditions). The binary cross-entropy (BCE) of the predicted SA events with respect to the ground truth is utilized, which is denoted as L_1. To better capture the RRI feature representation, we use FCN to cross-learn the RPA signal with the mean squared error (MSE) which is denoted as L_2. For learning the RPA feature representation, as learning the RRI feature information, leveraging the identical FCN to cross-learn the RRI signal with the MSE loss denoted as L_3. In this study, we utilize the joint learning scheme to learn the BAFNet model and its objective loss function can be defined as follows:

$$Loss = L_1 + L_2 + L_3 \tag{1}$$

2) RRI Stream Network: In order to extract the high-related feature information from an RRI segment, the RRI stream

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network is employed, which mainly consists of two modules: an encoder and a decoder. As shown in Table I, the first five layers belong to the encoder, and the rest five layers belong to the decoder. Specifically, for the encoder, we follow the backbone network in [14]. The input X^{RRI} first passes through two convolution layers with a max-pooling layer (Conv1) and obtains shallow-layer feature representations F_q^{RRI}, which is one of the inputs for the global query generation module. Then, a convolution layer with a max-pooling layer (Conv2) is followed to further learn the RRI feature representation, generating key vector K^{RRI} and value vector V^{RRI} for subsequent cross attention. Here, the value vector V^{RRI} is identical to the key vector K^{RRI}. Noted that the cross attention [46] $Att_{cross}^{RRI}(.)$ requires three inputs of K^{RRI}, global query vector Q, and V^{RRI}. Here, the global query vector is generated by the global query generation module, which is described in Section III-B4. Therefore, the RRI feature representation F^{RRI} can be learned and defined as follows:

$$F^{RRI} = Att_{cross}^{RRI}(Q, K^{RRI}, V^{RRI}) \tag{2}$$

Noted that the FCN achieved well performance on the task of image segmentation [47], [48], [49], helping the encoder much better learn the image feature representation. Similar to image segmentation, minimizing the loss between the RRI FCN-generated features and corresponding RPA can further help the RRI stream network to better learn the SA-related information. RRI and RAP are naturally correlated with each other as they are generated from the identical ECG signal. By computing the cross-loss between RRI and RAP, it will further enhance the BAFNet capability of learning the various feature information. Hence, for the decoder of the RRI stream network, we utilize the FCN to cross-learn the RPA signal. Firstly, F^{RRI} as the input of the decoder passes three transpose convolutional layers as well as an upsampling layer (TConv), output the F_{pre}^{RRI} which has the identical size with the input X^{RPA}. It can be defined to be:

$$F_{pre}^{RRI} = f^{RRI}(F^{RRI}) \tag{3}$$

where $f^{RRI}(\cdot)$ is the FCN in the RRI stream network.

3) RPA Stream Network: The RPA stream network has the identical architecture to the RRI stream network. Similar to the encoder in the RRI stream network, the feature representations obtained after Conv1 and Conv2 are called F_q^{RPA} and K^{RPA}, respectively. In addition, the copy of K^{RPA} is as the value vector V^{RPA}. After passing the cross-attention $Att_{cross}^{RPA}(\cdot)$, the feature F^{RPA} is obtained, which is defined as:

$$F^{RPA} = Att_{cross}^{RPA}(Q, K^{RPA}, V^{RPA}) \tag{4}$$

After F^{RPA} passes the decoder, the output reconstructs the RRI signal F_{pre}^{RPA}, which can be defined to be:

$$F_{pre}^{RPA} = f^{RPA}(F^{RPA}) \tag{5}$$

where $f^{RPA}(\cdot)$ is the FCN in the RPA stream network.

4) Global Query Generation: As shown in Fig. 2, there are RRI and RPA stream networks for learning feature representations for RRI and RPA signals. Cross-attention mechanism is a good way to learn the feature representations for RRI/RPA signals by controlling the information flow with attention. In this study, we propose a global query generation network with the input from both RRI and RPA signals. Specifically, the global query generation network includes two convolutional layers with a max-pooling layer (Conv2) and a self-attention block [50]. Here, we also call it bottleneck attention as [51], which can help RRI/RPA stream network to focus on high-related RRI/RPA information at a low computational cost. The input is the concatenation of F_q^{RRI} and F_q^{RPA} generated from Conv1 in the RRI and RPA stream networks. By passing the global query generation network $Att_{BN}(\cdot)$, the output of the global query vector Q can be obtained which is defined to be:

$$Q = Att_{BN}(F_q^{RRI}, F_q^{RPA}) \tag{6}$$

5) Feature Fusion: Fusing the feature representations from RRI and RPA stream networks is an important step to obtain well SA detection performance. Naïve concatenation or summation to fuse feature maps may limit the performance of the model to determine the relevant information [52], [53]. According to [14], attention is a well-presented operation to substitute naïve operation and adaptively fuse the feature maps. Therefore, channel-wise attention based on the squeeze-and-excitation block [54] is utilized here to learn the importance of feature maps extracted from different stream networks flexibly. As shown in Fig. 2, the input of the channel-wise fusion module is the stacking feature maps $F = [F^{RRI}, F^{RPA}]$. The channel-wise fusion module can assign different weights for different channels to explore valuable information. It includes squeeze, excitation as well as scale operations, and two main components are the squeeze block $f_{sq}(.)$ and excitation block $f_{ex}(.)$ to learn the importance λ of feature maps F. The channel-wise attention can be expressed as follows:

$$\lambda_i = f_{ex}(f_{sq}(F_i)) \tag{7}$$

$$F_i^{scale} = \lambda_i \cdot F_i \tag{8}$$

$$F^{scale} = [F_1^{scale}, F_2^{scale}, \ldots, F_C^{scale}] \tag{9}$$

where F^{scale} is the output feature maps of feature fusion module. $i \in [1, C]$ is the i-th channel, and C is the size of feature maps F.

C. Hard Sample Mining Scheme

In this study, there are two training stages. The first stage uses the entire training data for training to initially obtain a BAFNet model. In the second stage, only hard samples as the training data are utilized to refine the BAFNet model as hard sample scheme can well help the model discriminate the hard samples and improve the performance [39]. Here, we employ the clustering method of k-means for hard sample selection as the k-means method has the merits of simple, stable, and easy implementation. Here, we have to point out that only RRI signals are utilized for hard sample generating. The reason why we use RRI rather than RPA is discussed in Section V-H. Generally, a cluster often includes samples from different classes due to the indiscriminability of some samples. Therefore, we can

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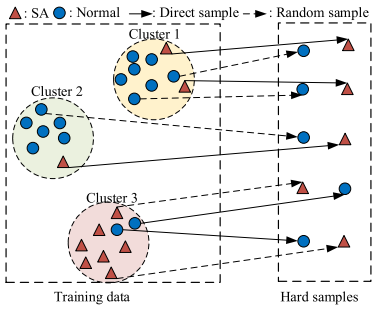


Fig. 3. Illustration of hard sample generation with k-means clustering. Minor samples are direct samples as hard samples as well as randomly selected major samples in a cluster.

generate hard samples according to this characteristic. Specifically, hard samples are generated by minor samples along with randomly selected major samples in a cluster. Here, the number of selected major samples is identical to the number of minor samples. Fig. 3 shows the detailed generation of hard samples.

IV. Experiment Setup

To obtain a well model generalization, there are some important hyperparameters to set up before training. Specifically, the training set is further split into training set and validation set at a ratio of 70%:30% in a stratified fashion. The maximum iterated epochs in the first training stage are set to 100. In the second training stage (*i.e.*, the refined training stage), the maximum iterated epochs are set to 5. The initial learning rate η is empirically set to 0.001. To accelerate the convergence speed, the learning rate keeps unchanged when the number of iterated epochs is less than or equal to 70. After over 70 iterated epochs, the learning rate is degraded with a factor of 0.1 per ten iterated epochs. The batch size is set to 128 to well balance the coverage and performance. The optimizer used in this study is Adam [55]. What's more, he_normal [56] is set to be the initializer of one-dimensional convolution layers and one-dimensional transpose convolutional layers. L_2 regularization is chosen to alleviate the overfitting problem. The rest of the hyperparameters are set to default values in the Keras framework.

V. Results and Discussion

A. Computing Environment

All experiments are conducted on a computing server equipped with an Intel(R) Xeon(R) CPU @ 2.20 GHz, Tesla P100-PCIE-16 GB, and 13.0 GBytes random access memory (RAM). The server installs a Linux system with the kernel version of 5.10.68. The proposed BAFNet is implemented based on the Keras framework at version 2.6.0 with the backend of TensorFlow 2.6.0.

B. Dataset

The dataset used for the experiments is the PhysioNet Apnea-ECG dataset [57], [58] which is available in the public domain.

TABLE II
Description of Recordings in PhysioNet Apnea-ECG Dataset

	Training set	Test set	Total
Number of subjects	35	35	70
Number of male subjects	30	27	57
Number of female subjects	5	8	13
Number of patient	23	23	46
Patient ratio	65.71 %	65.71 %	65.71 %

TABLE III
Description of Five-Minute-Long Segments in PhysioNet Apnea-ECG Dataset

	Training set	Test set	Total
Time duration (minutes)	16,709	16,945	33,654
Number of one-minute-long SA segments	6,473	6,490	12,963
SA ratio	38.74 %	38.30 %	38.52 %

There are 70 overnight recordings collected from 70 subjects (57 males, and 13 females), half of whom are divided into the training set and the rest of whom are in the test set in advance. The sampling frequency of ECG signals in this dataset is 100 Hz and the length of each ECG signal varies from slightly less than 7 hours to nearly 10 hours. On the basis of simultaneous respiration and oxygen saturation signals, experts derived a set of annotations for one-minute-long segments with apnea (A) and no apnea (N). Table II shows detailed information of the PhysioNet Apnea-ECG database and Table III lists the description of the five-minute-long segments driven from the PhysioNet Apnea-ECG database described in Section III-A. Moreover, the splitting of the training set and test set are recording-level, which means each overnight recording appears in the training set or test set but not both.

C. Performance Metrics

In this article, it includes two parts of performance metrics for per-segment classification and per-recording classification, which are described as follows:

1) Per-Segment Metrics: Accuracy (Acc), sensitivity ($Sens$), specificity ($Spec$), F_1 and area under the receiver operating characteristic curve (AUC) are utilized as the evaluation metrics for per-segment SA detection, which are defined to be:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \tag{10}$$

$$Sens = \frac{TP}{TP + FN} \tag{11}$$

$$Spec = \frac{TN}{TN + FP} \tag{12}$$

$$Prec = \frac{TP}{TP + FP} \tag{13}$$

$$F_1 = 2 \times \frac{Prec \times Sens}{Prec + Sens} \tag{14}$$

where TP represents the number of SA segments that are predicted as SA; TN denotes the number of normal segments that

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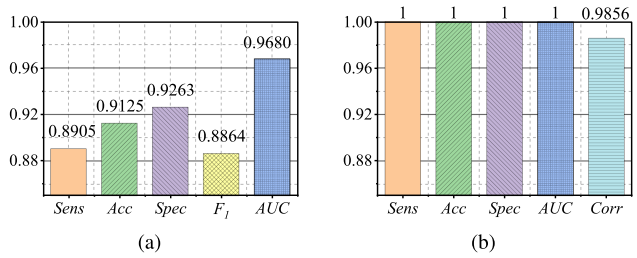


Fig. 4. Bar charts of SA detection performance on PhysioNet Apnea-ECG dataset: (a) per-segment; (b) per-recording.

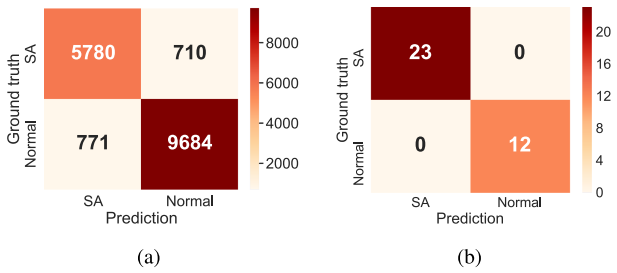


Fig. 5. Confusion matrix on PhysioNet Apnea-ECG datase: (a) per-segment; (b) per-recording.

are predicted as normal; FP represents the number of normal segments that are predicted as SA; FN denotes the number of SA segments that are predicted as normal.

2) Per-Recording Metrics: Per-recording metrics consist of accuracy, sensitivity, and specificity, as well as Pearson correlation coefficient ($Corr$). To obtain the metrics of accuracy, sensitivity, and specificity, it should define clearly whether an overnight recording is attributed to SA. According to the AASM guidelines, an overnight recording is SA when Apnea Hypopnea Index (AHI) is more than or equal to 5, otherwise normal. AHI is defined as follows:

$$AHI = \frac{60}{T} \times N \tag{15}$$

where T is the number of total one-minute-long segments in an overnight recording and N is the number of corresponding one-minute-long SA segments. Regarding the Pearson correlation coefficient, it is used to further evaluate the performance of the proposed method in per-recording SA detection, which is the correlation value between the ground truth and AHI.

D. Classification Performance

The proposed BAFNet is evaluated on the public PhysioNet Apnea-ECG dataset. The detailed SA detection performances of per-segment and per-recording classification are shown as follows:

1) Per-Segment Performance: As shown in Fig. 4(a), it is observed that the proposed BAFNet can achieve promising results, which can be up to 89.05%, 91.25%, 92.63%, 88.64%, and 96.80% in terms of sensitivity, accuracy, specificity, F_1 score and AUC value, respectively. Fig. 5(a) shows the confusion matrix, where we can note that most of the segments can be correctly identified.

2) Per-Recording Performance: As the per-recording performance is calculated based on the results of the per-segment classification. According to the definition of per-recording performance in Section V-C2, per-recording performance is obviously much better than the per-segment performance. As shown in Fig. 4(b), the BAFNet can achieve quite well per-recording results of 100% on performance metrics of accuracy, sensitivity, specificity, and AUC value. It also achieves 0.9856 $Corr$ which reflects the consistency of an overnight ECG recording between the ground truth AHI and predicted AHI. Fig. 5(b) shows the confusion matrix for the per-recording SA detection.

E. Visualization Analysis

As a deep neural network model like a black box, it cannot explain how the results come about. Class activation mapping (CAM) proposed by Zhou et al. [59] is a good strategy to alleviate this problem, due to the fact that it can clearly show us which part the model pays attention to. As reported in [60], [61], it is noted that CAM is often utilized for visual explanations in computer vision tasks. Therefore, in order to interpret the effectiveness of the proposed model more intuitively, CAM is applied to the RRI/RPA signals to show what is learned by the cross attentions in the RRI/RPA stream network.

Fig. 6 shows examples of cross attention with the global query in the RRI stream network. It is noted that the learned patterns of normal and SA segments are significantly different. As shown in Fig. 6(a) and (b), the importance distribution of cross-attention between normal and SA segments is significantly different. For example, the importance distribution for SA segments is almost located at the concave curves marked in red. While for normal segments, most concave curves are with less attention marked in blue.

As for the RPA stream network, cross attention also plays an important role in learning patterns from RPA signals. As shown in Fig. 7(a), it is observed that the cross attention assigns the weight importance almost the RPA signals for normal segments. Regarding the SA segments, Fig. 7(b) shows that the cross attention learns the weight importance which locates on some convex and concave curves. Therefore, the cross attention in the RPA stream network can well learn the high-related feature information among RPA signals.

F. Ablation Experiments

In order to verify the effectiveness of each component in the proposed BAFNet, we conduct a series of ablation experiments. Each experiment runs 10 times. Table IV shows the design of ablation experiments.

Table V shows the ablation experiment results. It is observed that the models of $M3$, $M4$, $M5$, and $M6$ with both RRI and RPA stream networks are obviously better than the models of $M1$ and $M2$ with a single RRI or RPA stream network. It means that the fusion of RRI and RPA signals can greatly facilitate the SA detection performance, at least improving 1.4%. As for the models of $M1$ and $M2$ with a single stream network,

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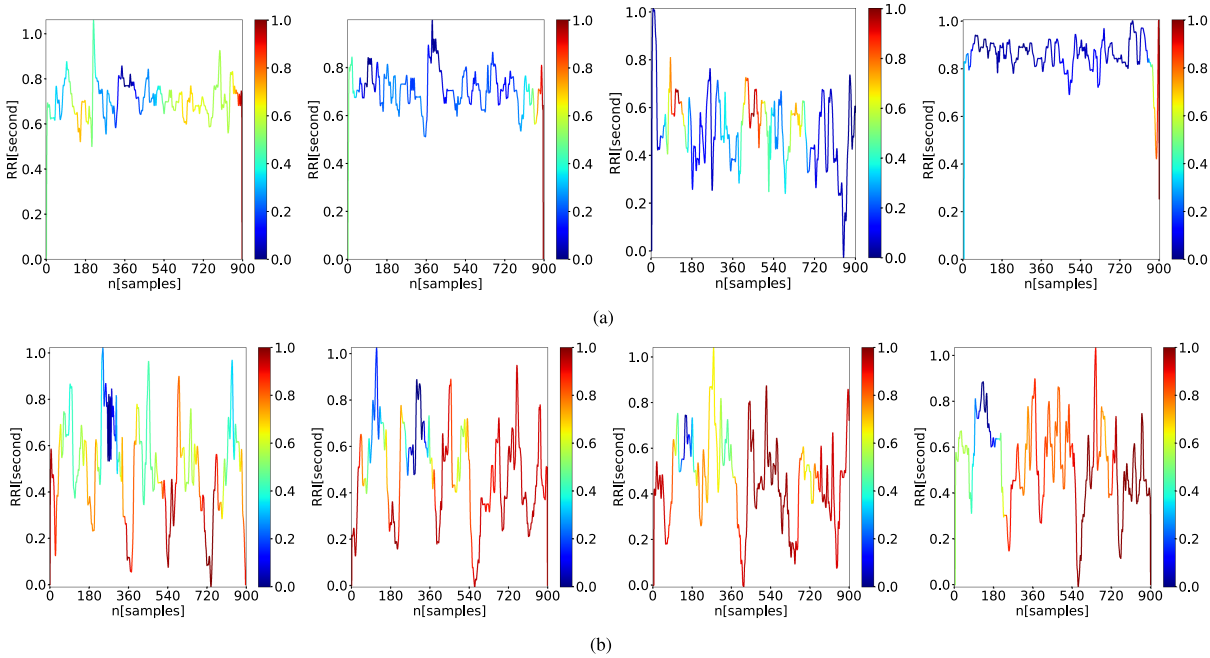


Fig. 6. Examples of the CAM for the cross-attention in the RRI stream network: (a) Normal; (b) SA.

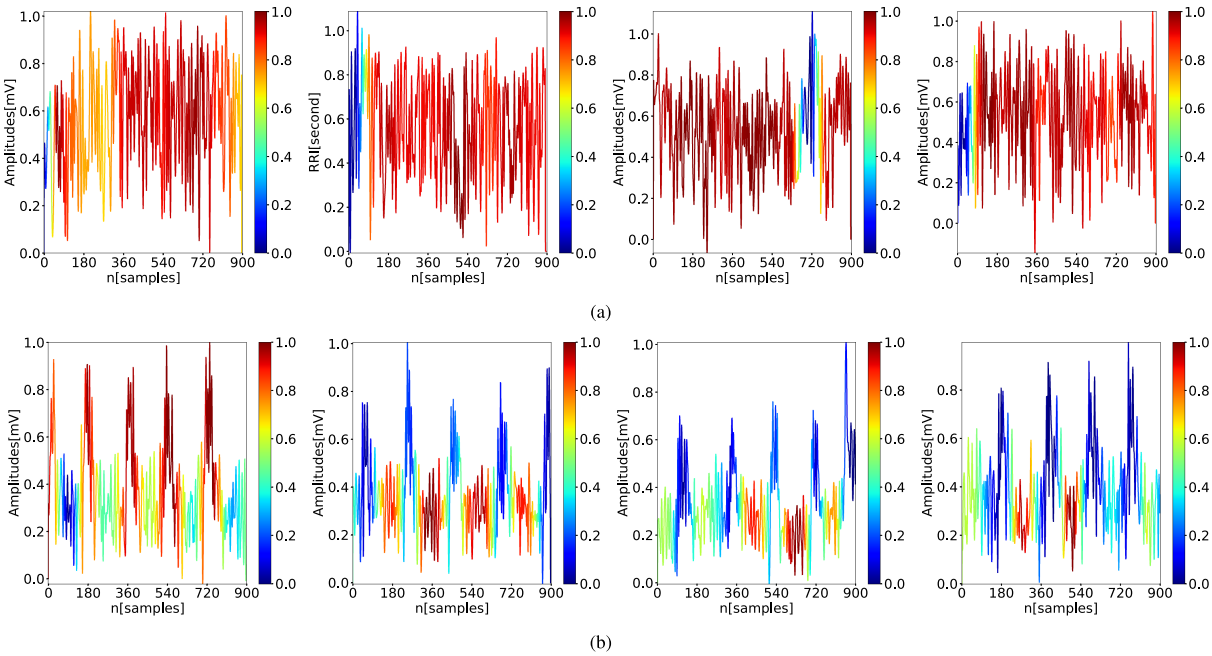


Fig. 7. Examples of the CAM for the cross-attention in the RPA stream network: (a) Normal; (b) SA.

their SA detection accuracy are comparable, but $M2$ achieves better sensitivity than $M1$ by approximately 4.9%. Compared with the model of $M3$, the model of $M4$ is with additional global query generation and its accuracy improves about 1.0%. It means that the global query generation is effective in the task of SA detection. As for the feature fusion, it can improve the robustness of the performance, especially for the accuracy. It also can help the model to obtain a promising performance. For example, compared to the model of $M4$, the model of $M6$ is with an additional feature fusion module. $M6$ achieves an accuracy of 0.9109 ± 0.0009 and $M4$ obtains an accuracy of 0.8913 ± 0.0042. The feature fusion module can improve about 2.0% accuracy as well as a smaller standard deviation. It means that the model with the feature fusion module is much better robust than the models without the feature fusion module.

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TABLE IV
THE DESIGN OF ABLATION EXPERIMENTS

Methods	RRI stream network	RPA stream network	Global query generation	Feature fusion
$M1$	✓			
$M2$		✓		
$M3$	✓	✓		
$M4$	✓	✓	✓	
$M5$	✓	✓		✓
$M6$	✓	✓	✓	✓

TABLE V
THE PERFORMANCE OF ABLATION EXPERIMENTS

Methods	*Acc*	*Sens*	*Spec*
$M1$	0.8441±0.0064	0.8113±0.0076	0.8645±0.0071
$M2$	0.8567±0.0093	0.8601±0.0087	0.8546±0.0113
$M3$	0.8829±0.0094	0.8825±0.0106	0.8832±0.0103
$M4$	0.8913±0.0042	0.8743±0.0143	0.9019±0.0062
$M5$	0.9006±0.0036	0.8864±0.0066	0.9094±0.0086
$M6$	**0.9109± 0.0009**	**0.8869±0.0175**	**0.9258±0.0115**

The bold values indicate the best performance.

TABLE VI
THE PERFORMANCE OF ABLATION EXPERIMENTS OF BAFNET WITH AND WITHOUT FCN

Model Settings	*Acc*	*Sens*	*Spec*
without FCN	0.8681	0.7575	0.9368
BAFNet	0.9109	0.8869	0.9258

Moreover, FCN is a part of RRI/RPA stream network, which can enhance the representation of features. To prove the effectiveness of FCN, we conduct an ablation experiment using BAFNet with and without FCN. As shown in Table VI, we can see that BAFNet with FCN can obviously achieve better accuracy and sensitivity even though the specificity decreases a little, which is about 1.1%. Specifically, BAFNet with FCN increases 4.28% accuracy and 12.94% sensitivity. It means that FCN can boost the BAFNet performance to some extent.

G. Performance Comparison

1) Per-Segment Classification: The per-segment classification performance comparisons with state-of-the-art SA detection methods are listed in Table VII. It is remarkable that our proposed BAFNet achieves competitive results for SA detection among all the cutting-edge methods. For the accuracy, our proposed BAFNet reaches up to 91.3%, 0.1% higher than the best existing state-of-the-art method [34]. Regarding the sensitivity, our proposed BAFNet is on par with WLTD [16], where the sensitivity is up to 89.0%. Compared with WLTD, our proposed BAFNet achieves much better accuracy and specificity. What's more, it is noted that CNN-BiGRU [62] has much better accuracy and specificity than ours. However, our proposed BAFNet achieves better sensitivity than CNN-BiGRU, where the sensitivity gap is up to 2.6%. Meanwhile, Table VIII, we can see that our proposed BAFNet is obviously superior to CNN-BiGRU on the per-recording performance. Furthermore, BAFNet has a smaller model size than CNN-BiGRU, which has a potential advantage in the applications of embedding in a portable device.

What's more, the methods [14], [62] that utilized adjacent segments as input always outperform the methods only using the target segments such as MSDA-1DCNN+WLTD [16] and ZF-Net+BiLSTM [63]. The accuracy of all the methods only using the target segments is under 90.0%. According to [14], the adjacent segments can improve the performance of the model due to the fact that the breathing condition of a person would not change all the time, which means in most situations, the adjacent segments have identical labels with the target segment. Therefore, when the target segment is contaminated by noise and the information it contains is insufficient, the adjacent segments would help greatly to some extent.

2) Per-Recording Performance: In addition to per-segment, our proposed BAFNet is also compared with the cutting-edge methods on the per-recording performance. As shown in Table VIII, it is observed that our proposed BAFNet achieves a comparable per-recording performance with the state-of-the-art methods. Even so, compared with the methods reported correlation coefficient, our proposed BAFNet achieves a little better than the best baseline. In addition, these state-of-the-art methods of [14], [16], [31], and [35] as well as our proposed BAFNet achieved all 100% in terms of accuracy, sensitivity, and specificity.

H. Hard Sample Analysis

In order to further improve the SA detection performance, we employ the hard mining technique described in Section III-C. Specifically, we use k-means to cluster the RRI signals for hard sample generation. Because it is quite significant between SA and normal RRI segments with $P = 1.58\text{E-}16$ ($\ll 0.001$). While it is not significant for RPA signals with $P = 0.284$ ($\gg 0.001$). Therefore, we merely utilize RRI signals for hard sample generations. To determine the number of clusters, we draw the loss curve of k-means and set the $k = 5$ at the elbow point (Fig. 8). By hard sample technique, the accuracy of BAFNet is improved from 90.92% to 91.25%.

I. Computational Complexity Analysis and Limitations

In this study, we calculate the total parameters of the proposed method which has 72,868 parameters and occupies $72,868 \times 4$ Bytes (≈ 0.28 MBytes). In the test phase, the proposed method only consumes 0.80 seconds for 35 overnight ECG records. That is to say, an overnight record with a length of 8.24 ± 0.52 hours only takes 0.02 seconds to obtain information on sleep quality. What's more, the proposed method takes up about 47.21 microseconds ($\frac{0.80\ seconds}{16945\ segments}$) to detect SA conditions for a piece of five-minute-long segment in per-segment classification. Due to the fact that the proposed method needs a post two

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TABLE VII
PER-SEGMENT PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS

Reference	Year	Feature extraction	Classifier	Input time length (min)	*Acc* (%)	*Sens* (%)	*Spec* (%)
Varon et al. [29]	2015	Feature engineering	LS-SVM	1	84.7	84.7	84.7
Song et al. [13]	2016	Feature engineering	HMM-SVM	1	86.2	82.6	88.4
Sharma et al. [64]	2016	Feature engineering	LS-SVM	1	83.4	79.5	88.4
Hassan et al. [30]	2017	Feature engineering	RUSBoost	1	88.9	87.6	91.5
Surrel et al. [65]	2018	Feature engineering	SVM	1	85.7	81.4	88.4
Li et al. [31]	2018	Auto-encoder	Decision fusion	1	84.7	88.9	82.1
Wang et al. [15]	2019	CNN	LeNet-5	5	87.6	83.1	90.3
Sharan et al. [66]	2020	CNN	1-D CNN	5	88.2	82.7	91.6
Feng et al. [67]	2020	Auto-encoder	TDCS	1	85.1	86.2	84.4
Bahrami et al. [63]	2021	CNN	LeNet+LSTM	1	80.7	75.0	84.1
Almutairi et al. [34]	2021	CNN+LSTM	SVM	5	90.9	91.2	90.4
Shen et al. [16]	2021	MSDA-1DCNN	WLTD	1	89.4	89.8	89.1
Chen et al. [14]	2021	CNN	SE-MSCNN	5	90.6	86.0	93.5
Yang et al. [35]	2022	CNN	1D-SEResGNet	2	90.3	87.6	91.9
Bahrami et al. [17]	2022	ZF-Net	BiLSTM	1	88.1	81.5	92.3
Chen et al. [62]	2022	CNN	CNN-BiGRU	5	91.2	86.4	94.1
This work	-	CNN	BAFNet	5	91.3	89.0	92.6

TABLE VIII
PER-RECORDING PERFORMANCE COMPARISON WITH STATE-OF-THE-ART METHODS

Reference	Year	*Acc* (%)	*Sens* (%)	*Spec* (%)	*Corr*
Song et al. [13]	2016	97.1	95.8	100	0.860
Sharma et al. [64]	2016	97.1	95.8	100	0.841
Li et al. [31]	2018	100	100	100	-
Wang et al. [15]	2019	97.1	100	91.7	0.943
Feng et al. [67]	2020	97.1	95.7	100	-
Shen et al. [16]	2021	100	100	100	-
Chen et al. [14]	2021	100	100	100	0.979
Yang et al. [35]	2022	100	100	100	0.985
Chen et al. [62]	2022	97.1	95.7	100	0.984
This work	-	100	100	100	0.986

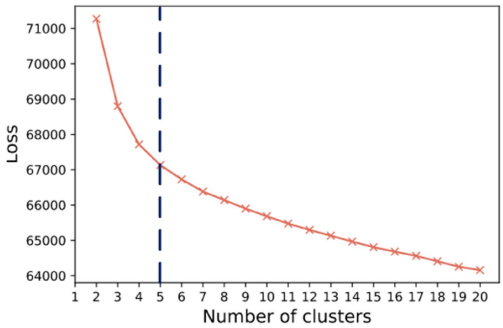


Fig. 8. Loss curve of k-means clustering. The elbow point is located at 5.

segments of the target as input, the prediction has a delay of about two minutes. It means that the user can get the results for each one-minute-long segment with two minutes input delay and 47.21 microseconds for SA detection.

However, BAFNet is trained and tested on the PhysioNet Apnea-ECG. The apnea-detection performance of BAFNet would decrease on an independent external dataset, which is a big challenge in the medical information field. Recently, large models, such as ChatGPT [68], SAM [69], and UniverSeg [70], achieved big progress when they were tested on an external dataset. In order to further improve the BAFNet generic performance, much more data should be collected and used for training, which is also the future work for obtaining a well generic BAFNet Model.

VI. CONCLUSION

In this study, a bottleneck attention based fusion network for sleep apnea detection named BAFNet is proposed. Specifically, BAFNet mainly consists of RRI stream network, RPA stream network, global query generation, feature fusion, and classifier. To alleviate the problem of insufficient information on a target segment, adjacent segments surrounding the target segment are combined together as input to be fed into BAFNet. To better learn the feature representation of RRI and RPA segments, FCN with cross-learning is employed. To better control feature information flow, a global query generation with bottleneck attention is proposed. To further improve the SA detection performance, a hard mining scheme with k-means clustering is employed. In addition, CAM of attention visualization is presented for interpretability. Experiment results on the PhysioNet Apnea-ECG dataset show that the proposed BAFNet can obtain promising per-segment performance and per-recording performance. The per-segment performance of BAFNet is 89.05%, 92.63%, and 91.25% in terms of sensitivity, specificity, and accuracy, respectively. As for per-recording performance, all sensitivity, specificity, and accuracy are 100%. Compared to the state-of-the-art SA detection methods, our proposed BAFNet achieves much better per-segment and per-recording performance. It means that our proposed BAFNet reaches the best among the state-of-the-art SA detection performance.

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